

Financial Derivatives

BBA — 8th Semester — 2021 — Answer Sheet
Subject Code: BNK202

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Section A

Brief Answer Questions: [10 × 1 = 10]

1. Financial derivatives are purely speculative, and highly leveraged instruments.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. If you buy a put option while holding the underlying stock, your maximum gain equals the stock purchase price minus the put premium minus the option striking price.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. Investors who do not consider risk in their decisions are said to be short selling.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Selling a call option is a bearish strategy that has limited gain and unlimited loss.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. There is inverse relation between interest rate and value of call.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. An option has an exercise price of Rs 25, the company declares a two for one stock split. Adjusted stock price would be Rs 50.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Put delta are negative, indicating the facts that the put option price and the underlying stock price are inversely related.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. In case of certainty, future price is simply the sum of spot price and cost of carry.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Swaps are custom-tailored to the need of the counterparties and often sold in organized exchange.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. In a currency swap, the notional principals are generally exchanged at the beginning and end of swap period.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

Section B

Short Answer Questions:[6 × 5 =30]

11. Discuss the current issues of derivative market in Nepal.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Consider the following option prices

Options	Call	Put
Strike price	100	100
Stock price	120	105
Option price	26	3.5

Risk-free interest rate is 6 percent per annum. Options expire in 90 days.
a. Which option is in-the-money and which is out-of-the-money?
b. Find the intrinsic value of options.
c. Find the time value of both options.
d. Assuming options are European, find the lower bound of both options.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

13. The stock price of Surya Hydro Company is Rs 140. A call option on this stock has exercise price of Rs 125 and selling for Rs 20. Another call option on this stock has Rs 130 exercise price and selling for Rs 12. Both calls are American and expire after 6 months. Explain whether you confirm the rules regarding American calls that differ only by exercise price. Also, suggest arbitrage strategy, if any.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

14. The stock of NMB bank sells for Rs 500 per share. There exists options on NMB stock, which permit its holder to buy a share at an exercise price of Rs 400. These options will expire after one year at which time stock will be selling either at Rs 800 or at Rs 200. Assume the risk free rate is 8 percent per annum.
a. Find the value of call option.
b. Show that how could you earn arbitrage profit if market price of call is Rs 220.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

15. Kathmandu Laboratories, Inc., is a new high-technology company whose common stock sells for Rs 230 per share. A call option exists on this stock with 3 months to expiration. It has an exercise price of Rs 215. You have made a careful study of the stock's volatility and conclude that a standard deviation of 0.50 is appropriate for the next 3 months. Currently, the risk-free interest rate is 6 percent per annum. Using the Black-Scholes-Merton option-pricing model, what is the fair price of European call on this stock?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

16. Currently the price of Chandragiri Hills stock is 2,500. You want to invest on this stock but you are not sure that whether the price of stock will increase or it will decrease in 6 months. Therefore, you want to make protective put. The put options with Rs 2,500 exercise price on Chandragiri stock are currently selling for Rs 500. a. How can you make protective put? b. What is the cost of protective put to you? c. At what stock price in 6 month will you breakeven? d. Calculate the gain/loss from protective put in 6 months if stock price of Chandragiri at that time turned to be Rs 1,000, Rs 1,500, Rs 2,000, Rs 2,500, Rs 3,000, Rs 3,500, Rs 4,000, or Rs 4,500 respectively and graph the results.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

17. A pension fund wants to enter into a six-month equity swap with a notional principal of Rs 60 million. Payments will occur in 90 and 180 days. The swap will allow the fund to receive the return on index 1, currently at 5514.67. The fund is considering three different types of swaps, one of which would require it to pay a fixed rate, another that would require it to pay floating rate, and another that would require it to pay the return on index 2, which is currently at 1212.98. Refer to these as swap 1, 2, and 3. The term structure is as follows: $\text{TermRateDiscount bond price} = \frac{1}{1 + \text{rate} \times \frac{\text{days}}{360}}$
 $90 \text{ days } 9\% B(90) = \frac{1}{1 + 0.09(90/360)} = 0.9780$
 $180 \text{ days } 10\% B(180) = \frac{1}{1 + 0.10(180/360)} = 0.9524$
a. Find the fixed rate for swap 1. b. Find the payments on day 90 for swaps 1, 2 and 3. Assume that on day 90 stock index 1 is at 5609.81 and stock index 2 is at 1231.94. Be sure to indicate the net payment. c. Assume it is 30 days into the life of swap. Stock index 1 is at 5499.62 and stock index 2 is at 1201.45. The new term structure is as follow: $\text{TermRateDiscount bond price} = \frac{1}{1 + \text{rate} \times \frac{\text{days}}{360}}$
 $60 \text{ days } 6.80\% B(60) = \frac{1}{1 + 0.068(60/360)} = 0.9888$
 $150 \text{ days } 7.05\% B(150) = \frac{1}{1 + 0.0705(150/360)} = 0.9715$
Find the value of swap 1, 2 and 3.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

18. A forward contract on silver has one year to expiration. The spot price of silver today is Rs 1,500 per Tola. It requires annual storage cost of Rs 50 per Tola payable in two installments after each six months. Interest rate is currently 8 percent per annum. One contract on silver covers 100 Tola. a. Find the forward price of silver today. b. Show arbitrage strategy if market price of forward contract is Rs 1,625. c. What is the initial value of forward contract? d. Assume it is 3 months after the contract was entered into. The spot price of silver now is Rs 1,350 per Tola and risk-free rate now is 8 percent per annum. What are the forward price and value of short forward contract now?

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.